MILITARY SPECIFICATION
FOR
ARRESTER, ELECTRICAL SURGE, F-1483( )/G and F-1495( )/G

This specification is approved for use by USA Communications Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers Arrester, Electrical Surge, F-1483( )/G (for roadside mounting) and F-1495/G (for curbside mounting).

1.2 Description.- The two arresters are designed to connect 12 4-wire channels from terminal electronic communication equipment to 12 4-wire channel field-wire or cable transmission lines and to isolate the terminal equipment from potentially damaging high voltage transients that may be carried in to the unit by the lines. The high-voltage transients from which protection is specifically required are those caused by electromagnetic pulse of high altitude nuclear detonations or by lightning. The F-1483/G arrester is designed for mounting in shelters with a roadside entry panel; the F-1495( )/G arrester is designed for mounting in shelters with a curbside entry panel.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.- The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications Research & Development Command, ATTN: DRDCO-CM-DM, Fort Monmouth, New Jersey 07703 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5920

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SPECIFICATIONS

MILITARY

MIL-S-45743	Soldering, Manual Type, High Reliability Electrical Electronic, Instrument, Communication, and Radar for Aerospace, and Control Systems, Procedures for
MIL-I-46058	Insulating Compound, Electrical
MIL-E-55585	Electronics Equipment and Parts

STANDARDS

MILITARY

MIL-STD-105	Standard Procedures and Tables for Inspection by Attributes
MIL-STD-726	Packaging Requirements Code
MIL-STD-810	Environmental Test Methods
MIL-STD-1458	Radioactive Materials: Marking and Labeling of Items, Packages and Shipping Containers for Identification in Use, Storage and Transportation.

DRAWINGS

USA CERCOM (FSCM 80063)

DL-SC-B-956111	Arrester, Electrical Surge F-1495( )/G
DL-SC-B-956112	Arrester, Electrical Surge F-1483( )/G

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENT

3.1 First article.- When specified in the contract or purchase order the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.- Arrester, Electrical Surge F-1483( )/G shall be constructed in accordance with drawings listed on DL-SC-B-956112; F-1495( )/G in accordance with drawings on DL-SC-B-956111.

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3.2.1 Marking.- The unit shall be marked in accordance with MIL-M-13231, MIL-STD-1458 and the drawings.

3.2.2 Serial numbers.- The arrester shall have consecutively assigned serial numbers unless otherwise specified by the contracting officer.

3.3 Performance requirements.-

3.3.1 Electrical performance.-

3.3.1.1 Static requirements.-

3.3.1.1.1 Sparkgap.- Breakdown of the sparkgap shall occur when a "fast rising pulse" of 1000 V or greater amplitude is incident on any of the input terminals (A through S). A fast rising pulse is defined for the purposes of this specification as one which rises from 10 percent of peak value in less than 20 nanoseconds and comes from a source resistance of 10 ohms or less. (See 4.6.1.1)

3.3.1.1.2 Breakdown voltage.- Application of a potential of 170 V DC, either polarity, between any pin (A through S) and the unit case shall result in a current flow greater than 1 MA DC. (See 4.6.1.2)

3.3.1.1.3 Bonds and Grounds.- Bonding and grounding impedances shall not exceed one milliohm. (See 4.6.1.3)

3.3.1.1.4 Continuity.- There shall be less than 50 ohms between any pair of input and output pins lettered A through S. (See 4.6.1.4)

3.3.1.1.5 Leakage.- Leakage current between any pin and the unit case shall not exceed 5 microamperes DC with an applied potential of 100 V DC, either polarity. (See 4.6.1.5)

3.3.1.2 Communication requirements.- The unit shall provide transmission paths for 12 4-wire channels with each path meeting the following requirements, excluding the interconnecting cables.

3.3.1.2.1 Transmission path impedance.- The input impedance for each transmission path when the output is terminated by 600 ohm ± 10 percent (at 16 and 32 kilobits per second) shall be as follows: (See 4.6.3)

Driving Signal	Maximum input Impedance
300 to 3500 Hz	600 ohms + 10 percent
16,000 Hz	750 ohms

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Driving Signal	Maximum input Impedance
32,000 Hz	750 ohms
64,000 Hz	750 ohms

3.3.1.2.2 Frequency response.- The frequency response for each transmission path through the unit (600 ohms ± 10 percent terminations) shall be as indicated: (See 4.6.4)

Frequency in Hz	Deviation from -4-dBm Input Reference
300 - 64,000	-1.0 dB to +0.5 dB

3.3.1.2.3 Distortion.- The total distortion produced by the transmission of any single test frequency between 300 and 64,000 Hz through the unit shall be at least 30 dB below the power of the test frequency. The test frequency shall be at a level of -4 dBm. (See 4.6.5.)

3.3.1.2.4 Signal-to-crosstalk ratio.- The signal-to-crosstalk ratio between any two terminal pairs in the unit (near-end and far-end) shall not be less than 60 dB at any frequency from 300 Hz to 64,000 Hz. (See 4.6.6.)

3.3.1.3 EMP protective requirement.- The unit shall reduce incoming high voltage transients due to EMP to 300 V peak at the unit output on any of the channels. For purposes of this specification, the transient due to EMP is defined as follows: (See 4.6.7.1.)

Minimum peak amplitude	12,000 V
Rise time, 10% to 90%	100 to 150 nanoseconds
Fall time, 100% to 5%	1 to 1.5 microseconds
Source impedance magnitude	20 to 24 ohms

3.3.1.4 Lightning protective requirement.- The unit shall reduce incoming high voltage transients due to lightning to 300 V peak at the unit output on any of the channels. For purposes of this specification, the transients due to lightning are defined as follows: (See 4.6.7.2.)

Minimum peak amplitude	20,000 V
Rise time, 10% to 90%	5 to 8 microseconds
Fall time, 100% to 5%	40 to 70 microseconds
Source impedance magnitude	6.0 ohms maximum

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3.3.2 Service environmental requirements.-

3.3.2.1 Temperature.- Unit performance and structural integrity shall not be adversely affected by temperature extremes (-70 to +145 °F) likely to be encountered in military service. (See 4.6.8 and 4.6.9)

3.3.2.2 Vibration and shock.- The unit performance or structural integrity shall not be degraded by vibration and shock incident to handling, shipment by ground or air transport, or service use. (See 4.6.10, 4.6.11, 4.6.12 and 4.6.13.)

3.3.2.3 Humidity.- Unit performance or sturctural integrity shall not be adversely affected by humidity conditions likely to be encountered in military service. (See 4.6.14)

3.3.2.4 Low pressure (altitude).- Unit performance or structural integrity shall not be adversely affected by reduced pressures due to shipment by air or installation at high elevations. (See 4.6.15.)

3.3.2.5 Salt fog.- Unit performance or integrity shall not be degraded by exposure of the unpackaged unit to a salt fog environment as may be encountered in service use. (See 4.6.16.)

3.3.2.6 Fungus.- Unit performance or structural integrity shall not be degraded by exposure of the unpackaged unit to conditions favorable to fungus attack, as may be encountered in service use. (See 4.6.17.)

3.4 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portion of the following paragraphs:

3.2 Construction
3.2.1 Marking
3.3 Performance requirments

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to

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perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities.- Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. Test equipment accuracy shall be taken into account in setting the acceptance limits. Standard and special test equipment is listed in Table I and II.

TABLE I Standard Test (STE) Equipment (Items listed or equivalent)

Designation	Function	Manufacturer
Simpson 1699	Milliohm meter	Simpson, Inc.
HP 200CD	Wide range oscillator	Hewlett-Packard Co.
HP 400D	Vacuum tube voltmeter	Hewlett-Packard Co.
TEK 485	Oscilloscope	Tektronix, Inc.
TEK P6015	Voltage probes	Tektronix, Inc.
HP 330B	Distortion meter	Hewlett-Packard Co.
260	Multimeter	Simpson
Type 1000SC	Package tester	L.A.B. Corp.
TEK 576	Curve tracer	Tektronix, Inc.

TABLE II Special Inspection Equipment (SIE)

Reference	Functional Description
Figure 1, 4.1.2.2.1	Pulse Generator, EMP
Figure 3, 4.1.2.2.2	Pulse Generator, Lightning
Figure 4, 4.1.2.2.3	Pulse Generator, Test
Figure 5, 4.1.2.2.4	Diode tester
Figure 6	Typical Test Connector

4.1.2.2.1 Pulse generator, EMP.- The EMP pulse generator (figure 1) produces a transient voltage that, for purposes of this specification, simulates the electromagnetic pulse (EMP due to a nuclear

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burst. The generator must be capable of repeated discharges into short circuit loads (50 milliohms) with a recycle time compatible with safety and a one-minute test cycle. Typically, the generator is connected between one input pin of test connector (figure 6) and ground. Upon triggering, the sparkgap discharges the energy storage device into the arrester (Figure 2) causing its protection sparkgap to fire. Subsequent output voltage, measured across a 600 ohm load at the pin of test connector corresponding to the input pin is significantly below the level of the input transient. After discharge, the energy storage device recharges automatically.

4.1.2.2.2 Pulse generator, lightning.- Operation and theory of the pulse generator, lightning (figure 3) is exactly the same as described in 4.1.2.2.1 with the one exception that the transient voltage more nearly simulates lightning.

4.1.2.2.3 Pulse generator, test.- This pulse generator (figure 4 provides a very low level pulse that induces an arc state in the protection spark gap to determine basic function capability. The test thus performed does not relate to either EMP or lightning, only to the functionality of the spark gap.

4.1.2.2.4 Diode tester.- Leakage and breakdown voltage can be measured using the diode tester (figure 5). Voltage is increased until reaching the proper level for test, and current is read while the button is depressed.

4.1.3 Standard conditions.- Tests shall be conducted at standard ambient conditions of temperature, pressure, and humidity as defined in MIL-STD-810, unless otherwise specified.

4.2 Classification of inspections.- Inspections shall be classified as follows:

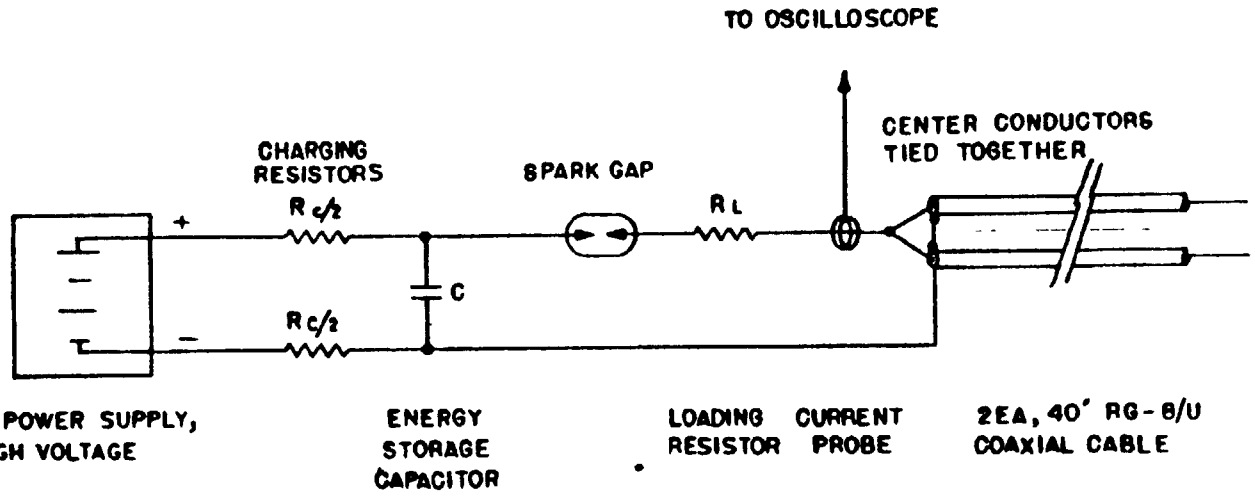
4.2.1 First article inspection
4.2.2 Quality conformance inspection.

4.3 First article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1.)

4.3.1 First article units.- The contractor shall furnish five (5) first article units of the complete arrester.

4.3.2 First article inspection.- The first article inspection shall consist of inspections specified in Table III and shall be performed in the order specified.

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1. POWER SUPPLY :	12 KV MINIMUM
2. CHARGING RESISTORS :	Rc = 2.5×10^8 Ohms MINIMUM
3. ENERGY STORAGE :	.02 μf 25 Kv, OIL FILLED
4 SPARK GAP, PRESSURIZED :	5 PSIG OF SF_6
5. LOADING RESISTOR :	10-12 Ohms
6. CURRENT PROBE :	SINGER 4M

FIGURE NO. 1
EMP GENERATOR

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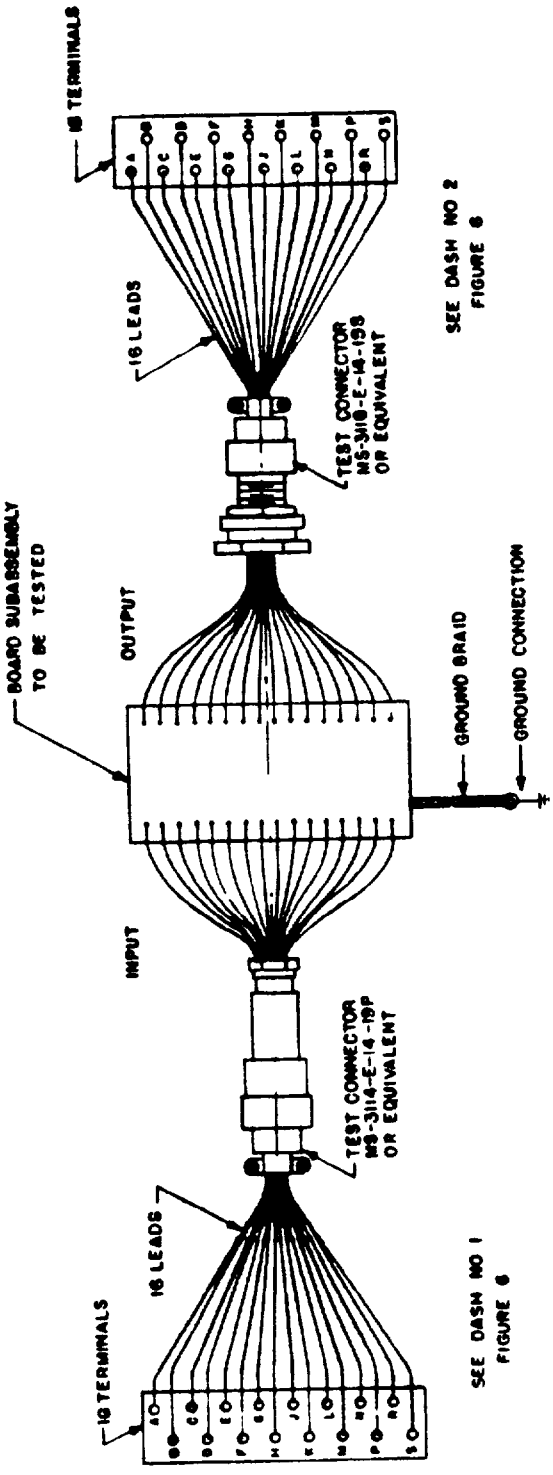


FIGURE NO 2
TPD ASSEMBLY
TEST ARRANGEMENT

NOTES
1 TEST CONNECTORS MATED TO APPROPRIATE TPD ASSEMBLY CONNECTORS AND GROUND BRAID ELECTRICALLY CONNECTED

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4.3.3 First article data.- The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Quality conformance inspection.- Quality conformance inspection shall consist of Groups A and B inspections (Tables IV and V) and Material. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.4.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table IV. Lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table IV.

4.4.2 Lot formation.- After assembly and Group A tests, discrete lots shall be assembled comprising one weeks production or 200 units, whichever is least.

4.4.3 Sampling.- Sampling for Group B tests shall be in accordance with MIL-STD-105, inspection level S-4. For Group C tests, two units a month shall be selected for tests, and an additional unit shall be selected every three months.

4.5 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspecton requirements specified in the contract or purchase order.

Item	Where Required
Marking	3.2.1

4.6 Test Methods.-

4.6.1 Arrester static electrical tests.- (Figure 2 applies.)

4.6.1.1 Sparkgap.- A fast rising test pulse generator (figure 4, para 4.1.2.2.3, or equivalent) shall be used to verify sparkgap arcing. Sparkgap voltage measure at the input terminals

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TABLE III First Article Inspection

Inspections	Reqt Para	Test Para	Test Sequence Unit Number 1	2	3	4	5
Inspection covered by Subsidiary Documents		4.5	Inspect all units				
Bounce preconditioning		4.6.2	Perform on all units				
Group A Inspect	See Table IV		Inspect all units				
Group C Inspect	See Table V		Inspect all units				
Subgroup 1							
EMP	3.3.1.3	4.6.7.1	1	1	1	1	1
Lightning	3.3.1.4	4.6.7.2	2	2	2	2	2
Subgroup 2							
High Temp	3.3.2.1	4.6.8		3			
Low Temp	3.3.2.1	4.6.9		4			
Subgroup 3							
Vibra/Shock	3.3.2.2	4.6.10			3		
		4.6.11			4		
		4.6.12			5		
		4.6.13			6		
Subgroup 4							
Humidity	3.3.2.3	4.6.14	3				
Low pressure	3.3.2.4	4.6.15		5			
Salt fog	3.3.2.5	4.6.16*					3
Fungus	3.3.2.6	4.6.17*				3	

* The unit shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

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TABLE IV Group A Inspection

Inspections	Reqt Para	Test Para	Test Equip	AQL/%DEF Major	AQL/%DEF Minor
				(SEE NOTE)	
Circuit board/busbar assembly					
Minor defects					
201 Warpage of board or busbar after soldering.		Visual			1.0
202 Solder connection filletted improperly.	**	Visual			1.0
Wiring harness					
Major defect					
101 Wire color at pin		Visual		1.0	
102 Solder points	**	Visual		1.0	
Minor defects					
201 Greater than 3 broken wire strands	*	Visual			1.0
202 Shrink Tube sleeves		Visual			1.0
203 Rubber insulating insert		Visual			1.0
Circuit board assembly					
101 Solder connections		MIL-S-45743		1.0	
102 Component & Wire clearance		Visual		1.0	
103 Connector hardware missing or loose		Visual		1.0	
104 Diode polarity		Visual		1.0	

*Rework required if greater than 3 broken strands
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TABLE IV Group A Inspection (cont'd)

Inspections	Reqt Para	Test Para	Test Equip	AQL/%DEF Major	AQL/%DEF Minor
					(See Note)
105 Sparkgap ground resistance		4.6.1.3		1.0	
106 Input wire resistance		4.6.1.3		1.0	
Minor defects					
201 Wire lacing		Visual			1.0
202 Ground lug, excess solder & braid		Visual			1.0
203 Conformal coating	***	Visual			1.0
***MIL-I-46058					
Case, cover					
Major defects					
101 Plating		Visual		1.0	
102 Case-to-cover fit		Visual		1.0	
103 Welds		Visual		1.0	
Minor defects					
201 General cleanliness		Visual		1.0	
Case with circuit subassemblies					
Major defects					
101 Spacers and hardware lugs		Visual		1.0	
102 Output receptacles resistance 2 case		4.6.1.3		1.0	
103 Ground braid resistance		4.6.1.3		1.0	
Minor defects					
201 Placement of wiring		Visual			1.0

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TABLE IV Group A Inspection (cont'd)

Inspections	Reqt Para	Test Para	Test Equip	AQL/%DEF Major	Minor
202 General cleanliess		Visual			1.0
Arrester assembly					
Major defects					
101 Hardware screw missing or loose		Visual		1.0	
Minor defects					
201 Nameplate missing		Visual			1.0
Performance tests					
Sparkgap	3.3.1.1.1	4.6.1.1	SIE	1.0	
Breakdown voltage	3.3.1.1.2	4.6.1.2	SIE	1.0	
Bonds & grounds	3.3.1.1.3	4.6.1.3	STE	1.0	
Continuity	3.3.1.1.4	4.6.1.4	SIE/STE	1.0	
Leakage	3.3.1.1.5	4.6.1.5	SIE	1.0	

Note: Defects per hundred units

TABLE V Group B Inspection

Inspection	Reqt Para	Test Para	Test Equip	AQL Major	Minor
Arrester assembly					
Electrical Tests					
Channel impedance	3.3.1.2.1	4.6.3	STE	1.0	
Frequency	3.3.1.2.2	4.6.4	STE	1.0	
Distortion	3.3.1.2.3	4.6.5	STE	1.0	
Signal-to-Crosstalk	3.3.1.2.4	4.6.6	STE	1.0	

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TABLE VI Group C Inspection

Inspections	Reqt Para	Test Para	Test Equip	AQL Major	AQL Minor
Subgroup 1					
EMP	3.3.1.3	4.6.7.1	SIE	1.0	
Lightning	3.3.1.4	4.6.7.2	SIE	1.0	
Subgroup 2					
High Temp	3.3.2.1	4.6.8	STE	1.0	
Low Temp	3.3.2.1	4.6.9	STE	1.0	
Subgroup 3					
Vibra/Shock	3.3.2.2	4.6.10	STE	1.0	
	3.3.2.2	4.6.11	STE	1.0	
	3.3.2.2	4.6.12	STE	1.0	
	3.3.2.2	4.6.13	STE	1.0	
Subgroup 4					
Humidity	3.3.2.3	4.6.14	STE	1.0	
Low pressure	3.3.2.4	4.6.15	STE	1.0	
Salt Fog	3.3.2.5	4.6.16*	STE	1.0	
Fungus	3.3.2.6	4.6.17*	STE	1.0	

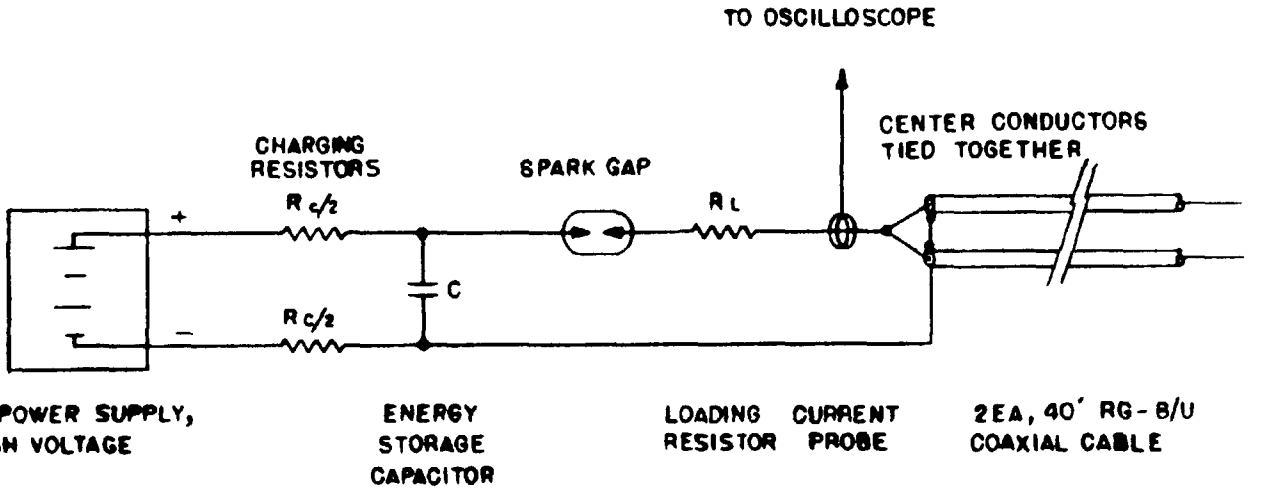
*The unit shall be throughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

of the arrester shall drop to less than 100 volts in a precipitous fashion on application of the pules of requirements para 3.3.1.1.1. The appropriate test connector shall be used (Figure 6. The requirements of 3.3.1.1.1 shall be met.

4.6.1.2 Breakdown voltage.- The direct current breakdown shall be measured to ground from all input and output pins (A through S) using the SIE. See figure 5 for applications of SIE. The requirements of 3.3.1.1.2 shall be met.

4.6.1.3 Bonds and grounds.- Component connections, ground bus connections, ground braid connections and ground lug connections shall be measured for low resistance using a milliohmmeter, part of the STE. The requirement of 3.3.1.1.3 shall be met.

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1 POWER SUPPLY :	20 KV MINIMUM
2 CHARGING RESISTORS :	$Rc = 2 \times 10^5$ Ohms MINIMUM
3. ENERGY STORAGE :	16 μf 25 Kv, OIL FILLED (OR EQUAL)
4 SPARK GAP, PRESSURIZED :	5 PSIG OF SF_6
5 LOADING RESISTOR :	3.0 Ohms MAXIMUM
6 CURRENT PROBE :	SINGER 4M

FIGURE NO. 3
LIGHTNING PULSE GENERATOR

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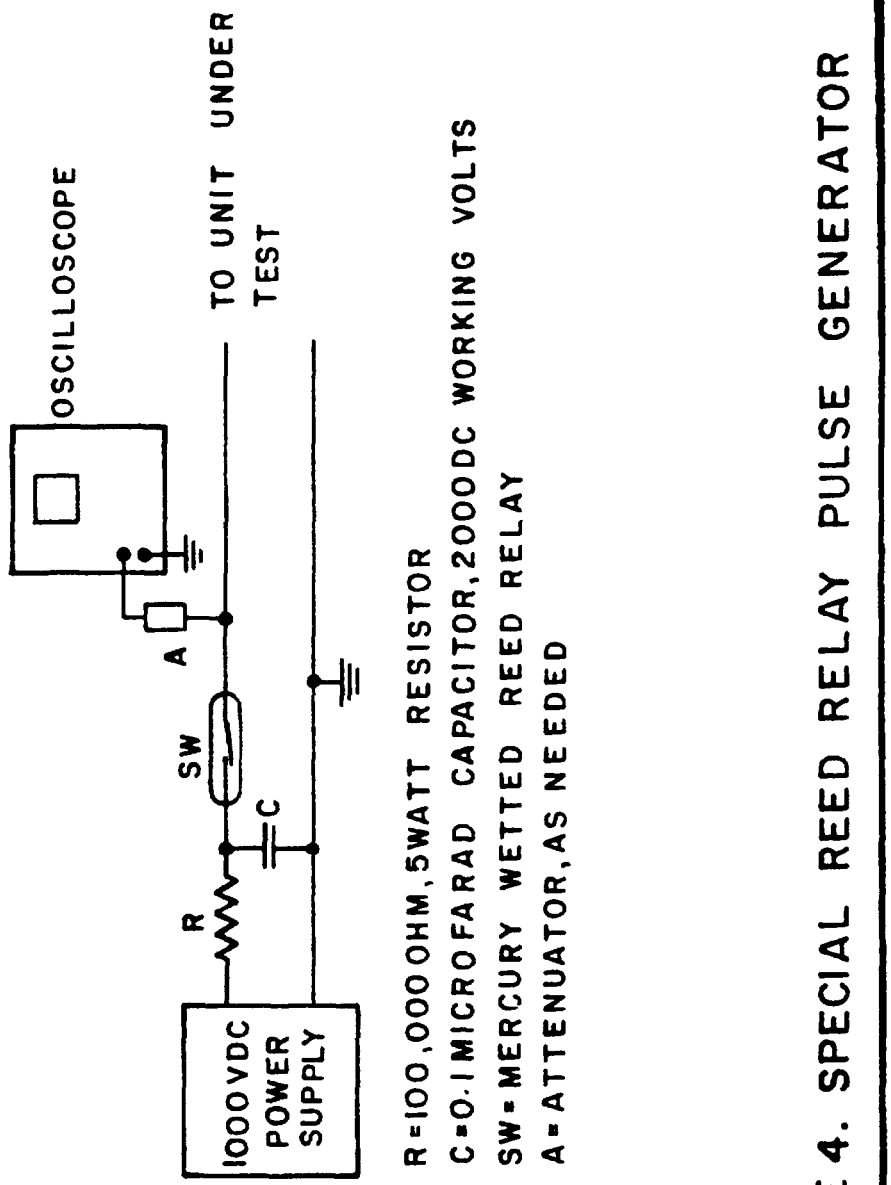
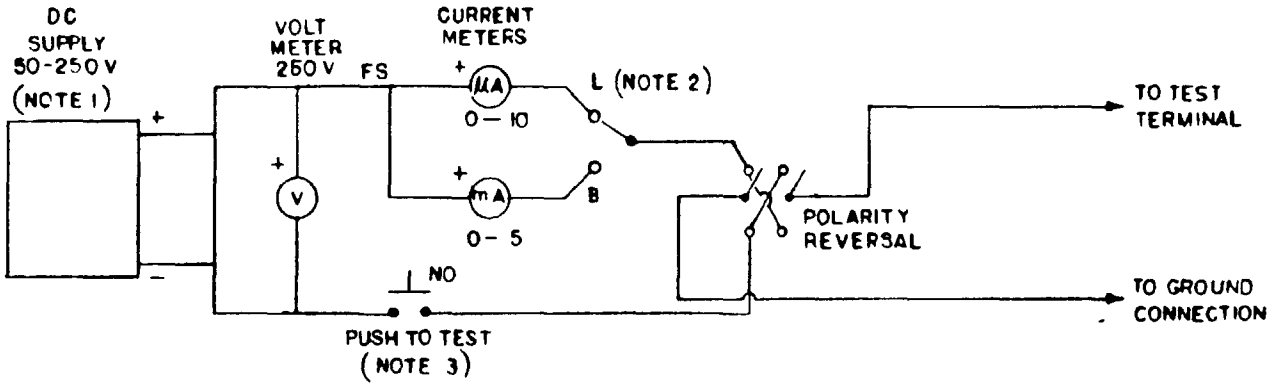


FIGURE 4. SPECIAL REED RELAY PULSE GENERATOR

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NOTE 1 CURRENT LIMITED DC SUPPLY, 10 mA MAX , VOLTAGE CONTINUOUSLY VARIABLE

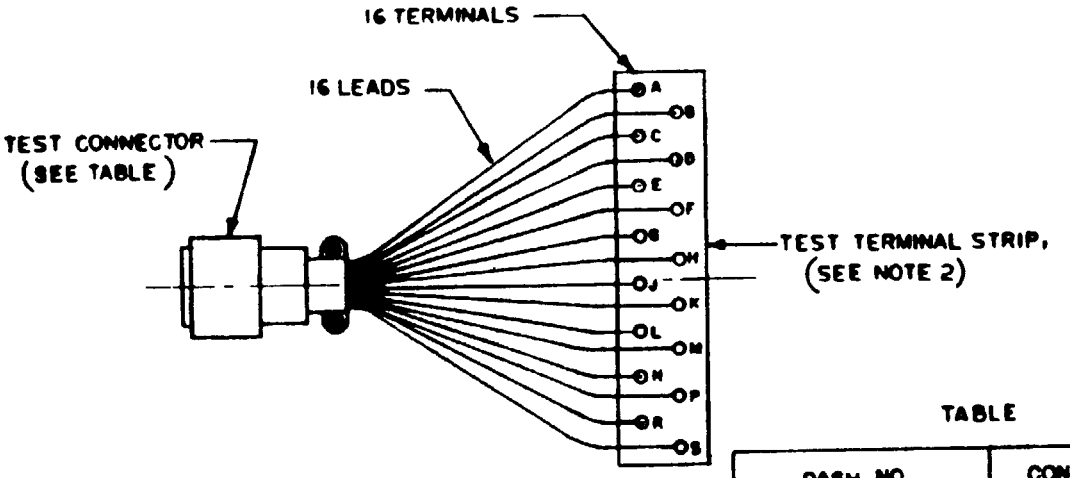
NOTE 2: "L" POSITION FOR LEAKAGE TEST, 5μA POINT MARKED 0-10 μA METER;
"B" POSITION FOR BREAKDOWN TEST, 1 mA POINT MARKED ON 0-5 mA METER.

NOTE 3 NORMALLY OPEN PUSHBUTTON CLOSED TO TEST.

NOTE 4 EQUIVALENT APPARATUS MAY BE USED TO DEMONSTRATE SATISFACTION OF 3.3.1.1 STATIC CHARACTERISTICS OF REQUIREMENTS SECTION.

FIGURE NO. 5
SCHEMATIC DIAGRAM
DIODE TESTER

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TABLE

DASH NO.	CONNECTOR NO.
1	MS-3114-E-14-19P
2	MS-3116-E-14-19S

NOTES

1. TEST VOLTAGE TO BE APPLIED BETWEEN TEST TERMINALS AND GROUND IN BOTH POLARITIES USING TPD ASSEMBLY TEST APPARATUS
2. LETTER DESIGNATION CORRESPONDS TO CONNECTOR PINS

FIGURE NO. 6
TYPICAL TEST CONNECTOR

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4.6.1.4 Continuity.- The direct current resistance from each pin (A through S) at the input side of the arrester shall be measured to the appropriate pin at the output side using STE. See figure 2 for typical test connections and application of SIE. The requirement of 3.3.1.1.4 shall be met.

4.6.1.5 Leakage.- The direct leakage current shall be measured to ground from all input and output pins using the SIE. See figure 5 for applications of SIE. The requirements of 3.3.1.1.5 shall be met.

4.6.2 Bounce preconditioning.- The unpackaged unit shall be placed in its normal operating position on the table of a package tester (type 1000 SC, L.A.B. Corporation, Skaneatales, New York, or equivalent). The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. Each unit of each lot shall be subjected to this preconditioning. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to being subjected to quality conformance inspections.

4.6.3 Transmission path impedance test.- The unit shall be connected as shown in figure 7. The impedance of each channel shall be measured at 300, 1000, 3500, 16,000, 32,000, and 64,000 Hz. The requirement of 3.3.1.2.1 shall be met.

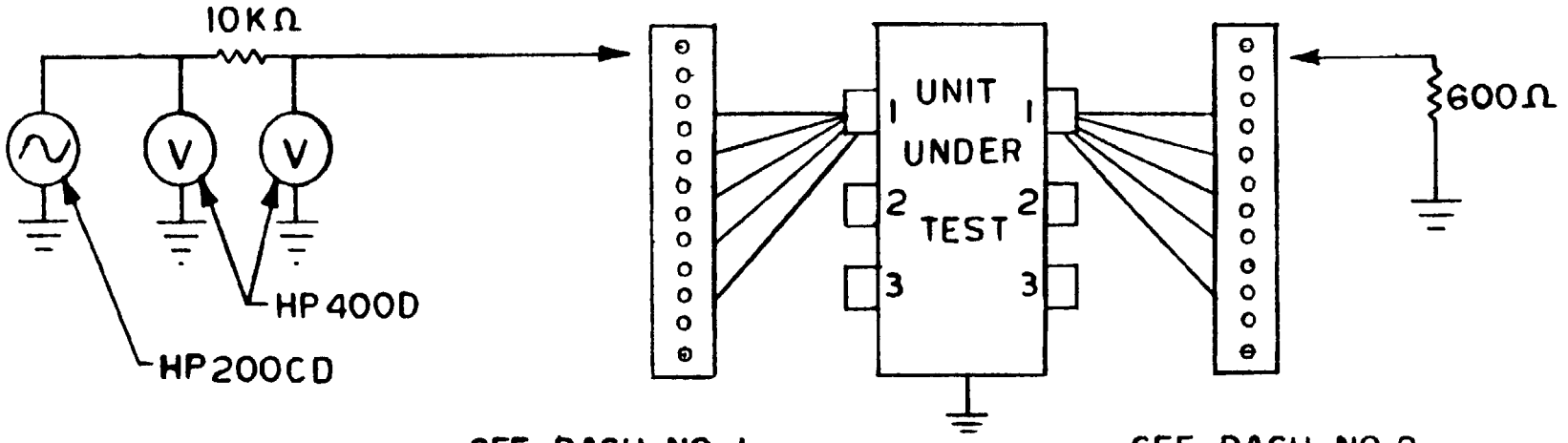
4.6.4 Frequency response test.- The unit shall be connected as shown in figure 8. The frequency response of all channels shall be measured at 300, 1000, 3500, 16,000, 32,000, and 64,000 Hz. The requirement of 3.3.1.2.2 shall be met.

4.6.5 Distortion test.- The unit shall be connected as shown in figure 9. The total distortion of output signals of all channels shall be measured at frequencies of 300, 1000, 3500, 16,000, 32,000 and 64,000 Hz. The requirement of 3.3.1.2.3 shall be met.

4.6.6 Signal-to-crosstalk test.- The unit shall be connected as shown in Figure 10. The signal to crosstalk plus noise from channels 1, 5, and 9 to all other channels shall be measured at 300, 1000, 3500, 16,000, 32,000 and 64,000 Hz. The requirement of 3.3.1.2.4 shall be met.

4.6.7 Operational tests.- Connect the unit, pulse generator and related test equipment as shown in figure 11. EMP tests shall be conducted using the EMP pulse generator; lightning tests shall be conducted using the lightning pulse generator.

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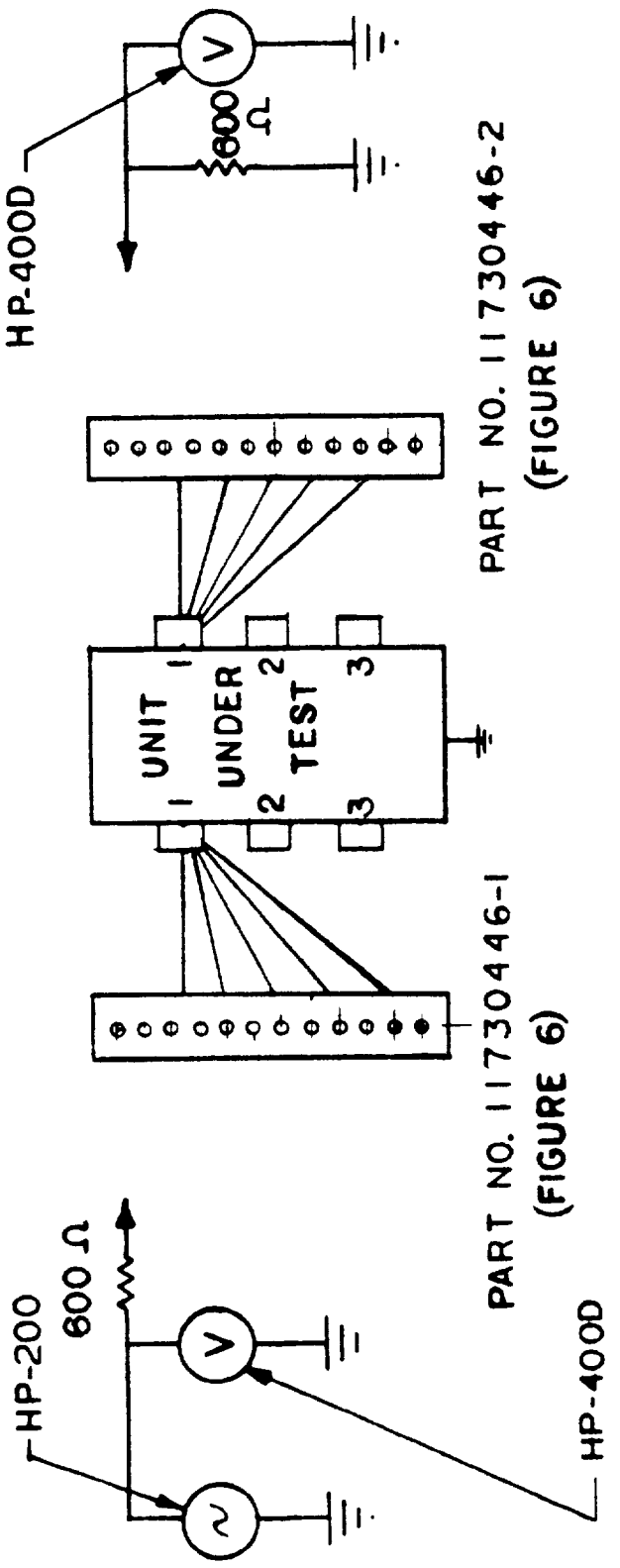
SEE DASH NO. 1
FIGURE 6

SEE DASH NO. 2
FIGURE 6

REPEAT TEST FOR CONNECTORS 2 & 3 ON UNIT UNDER TEST

FIGURE 7: EQUIPMENT INTERCONNECTION FOR MEASURING CHANNEL IMPEDANCE.

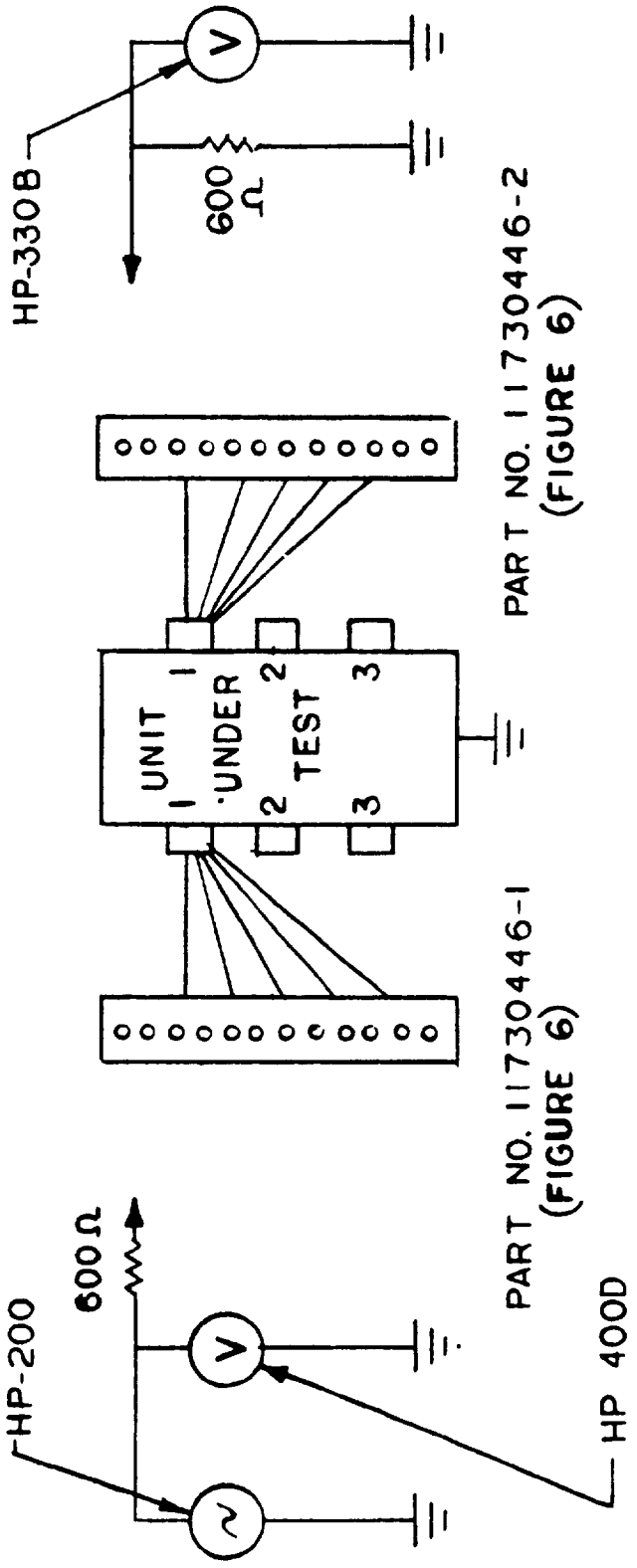
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REPEAT TEST FOR CONNECTORS 2 & 3 ON UNIT UNDER TEST.

FIGURE 8: EQUIPMENT INTERCONNECTION FOR MEASURING FREQUENCY RESPONSE

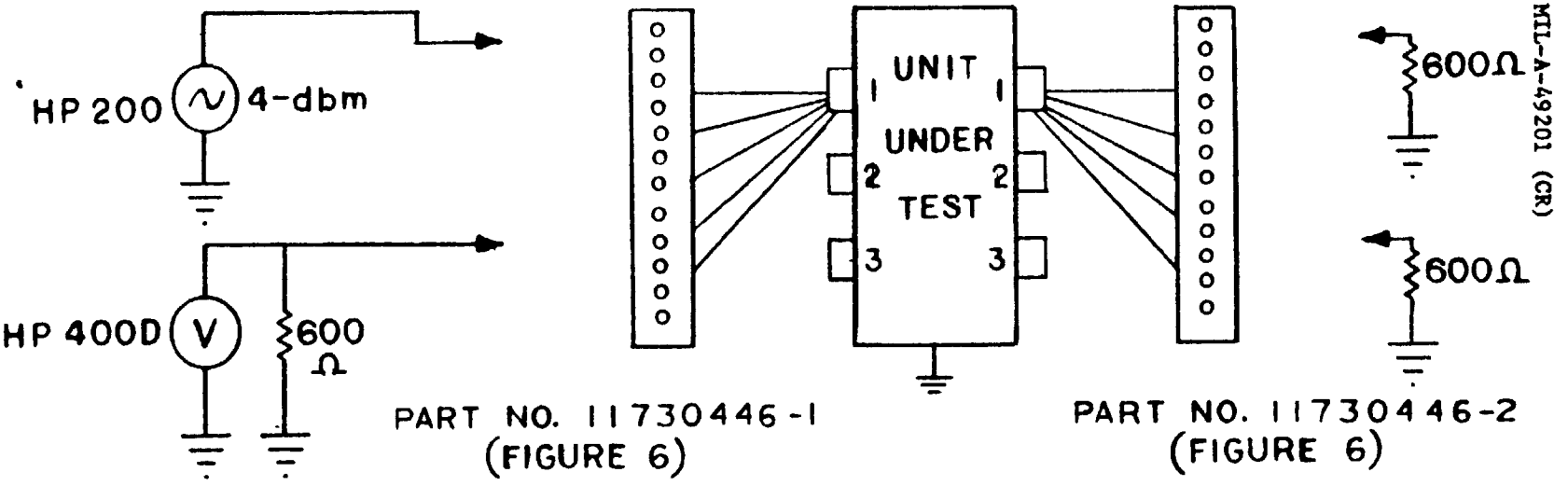
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REPEAT TEST FOR CONNECTORS 2 & 3 ON UNIT UNDER TEST.

FIGURE 9: EQUIPMENT INTERCONNECTION FOR MEASURING DISTORTION

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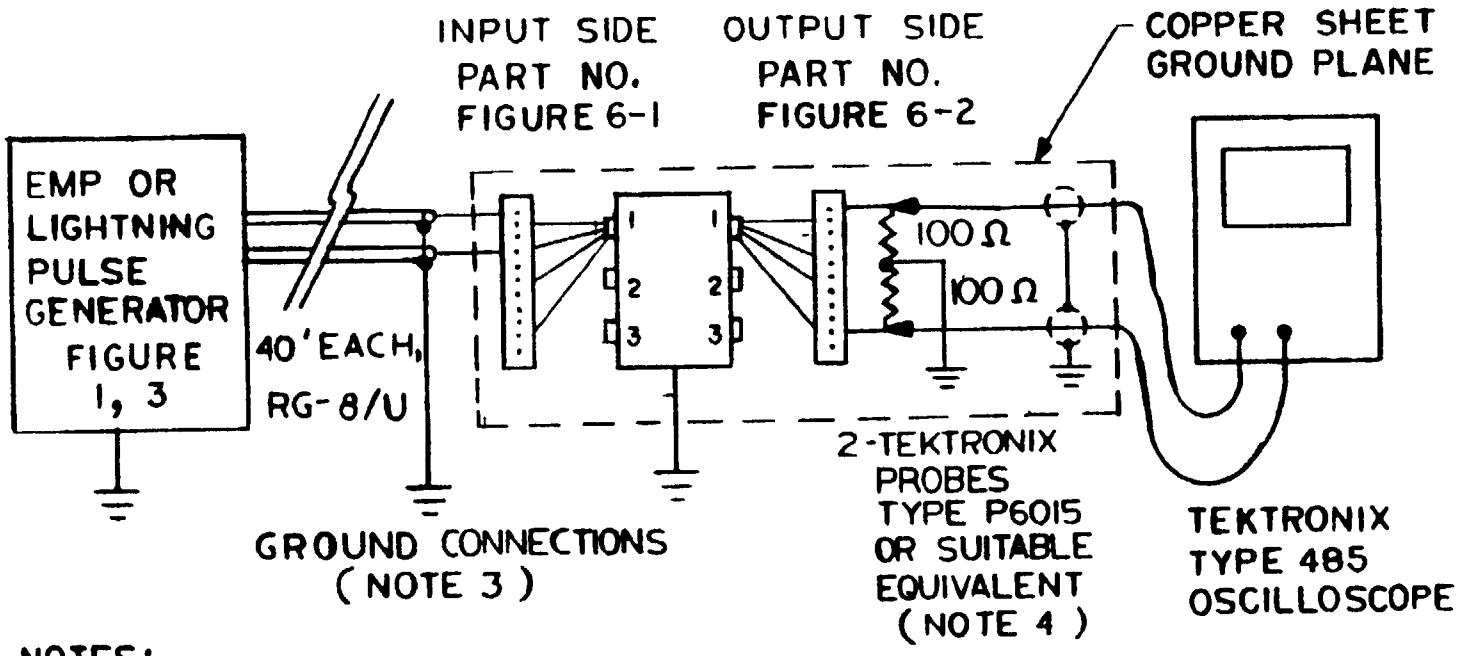
PART NO. 11730446-1
(FIGURE 6)

PART NO. 11730446-2
(FIGURE 6)

REPEAT TEST FOR CONNECTORS 2 & 3 ON UNIT UNDER TEST

FIGURE 10: EQUIPMENT INTERCONNECTION FOR MEASURING CROSSTALK

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NOTES:

1. FIGURE 6-1, 2 CONNECTED TO INPUT AND OUTPUT CONNECTOR PAIRS WHICH ARE INTERNALLY CONTINUOUS
2. FIGURE 6-1, 2 ARE MOVED TO NEXT SET OF CONNECTORS AFTER ALL CHANNELS ON PREVIOUS SET HAVE BEEN TESTED UNTIL ALL CHANNELS ARE TESTED.
3. CAREFUL GROUNDING MAY BE REQUIRED TO REDUCE NOISE PICKUP
4. SHIELDED PROBES MAY BE REQUIRED TO REDUCE NOISE PICKUP.

FIGURE 11: EQUIPMENT INTERCONNECTION FOR OPERATIONAL TESTS

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4.6.7.1 EMP test.- Apply three pulses to the input of each two wire channel and observe the output pulses on each of the appropriate output leads using the two oscilloscope channels independently. The requirements of 3.3.1.3 shall be met. Repeat for all channels of unit.

4.6.7.2 Lightning test.- Apply three pulses to the input of each two-wire channel and observe the output pulses on each of the appropriate output leads using the two oscilloscope channels independently. The requirements of 3.3.1.4 shall be met. Repeat for all channels of unit.

4.6.8 High temperature.- The unit shall be conditioned in accordance with MIL-STD-810, Method 501.1, Procedure II, except the unit shall be subjected to three 12-hour temperature cycles. The assembly shall pass Group A and B electrical tests during and after the test as specified in method 501.1. (See 3.3.2.1.)

4.6.9 Low temperature.- The unit shall be conditioned in accordance with MIL-STD-810, Method 502.1, Procedure I. The chamber temperature shall be lowered at a rate not exceeding 10°C (18°F) per minute. The unit shall be subjected to Group A and B electrical tests during and after conditioning as specified in method 502.1. (See 3.3.2.1.)

4.6.10 Vibration test.- The unit shall be tested in accordance with MIL-STD-810, Method 514.2, Procedure VIII, in accordance with schedule of Table 514-2VI for wheeled vehicles for the maximum time (5-1/2 hours each axis). Displacement shall be in accordance with Curve V of figure 514.2-6. The unit shall pass Group A and B Electrical tests for the full assembly on completion of conditioning. (See 3.3.2.2.)

4.6.11 Shock (basic design) test.- The unit shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure I. The shock pulse shall be in accordance with figure 516.2-1 with a peak value of 20 g. Unit shall pass Group A and B electrical tests for the full assembly on completion of conditioning. (See 3.3.2.2)

4.6.12 Shock (bench handling) test.- The unit shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure V. After conditioning, the unit shall pass Group A and B electrical tests for the full assembly. (See 3.3.2.2)

4.6.13 Bounce (loose cargo) test.- The packaged unit shall be tested in accordance with MIL-STD-810, Method 514.2, Procedure XI, Part 2. After conditioning, the unit shall pass Group A and B electrical tests for the full assembly. (See 3.3.2.2.)

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4.6.14 Humidity test.- The unit shall be tested in accordance to MIL-STD-810, Method 507.1, Procedure II except the unit shall be submitted and shall pass Group A and Group B visual and electrical tests for the full assembly during and after conditioning has been completed (Step 7). (See 3.3.2.3)

4.6.15 Low pressure (altitude) test.- The unit shall be tested in accordance with MIL-STD-810, Method 500.1, Procedure I. The unit shall pass Group A and B electrical tests for the full assembly at reduced pressure and at the end of conditioning in accordance with Procedure I. (See 3.3.2.4)

4.6.16 Salt fog test.- The unpackaged unit shall be tested in accordance with MIL-STD-810, Method 509.1 except that after salt fog conditioning, the unit shall be dryed 48 hours at ambient conditions and shall then pass Group A and B visual and electrical tests for the full assembly. (See 3.3.2.5)

4.6.17 Fungus test.- The unpackaged unit shall be tested in accordance with MIL-STD-810, Method 508.1. On completion of conditioning, the unit shall be inspected in accordance with Procedure I, with Group A and B visual and electrical tests for the full assembly applicable. (See 3.3.2.6)

5. PACKAGING

5.1 Preservation.- Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A.- 1Ø-1-1-ØØ-OO-NA-X-B7-A-ED-A.

5.1.2 Level B.- 1Ø-1-1-ØØ-OO-NA-X-B7-A-ED-B.

5.2 Packing and marking.- Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use.- The intended use of this device is to protect field communication equipment from electrical surge current due to EMP or lightning. The specific applications are classified information.

6.2 Ordering information.-

6.2.1 Documents.- Procurement documents should specify the title, date and number of this specification.

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6.2.2 Government testing of comparison samples.- These samples are subjected to retesting by the Government to insure that the item continues to pass the first article inspection. Failure of these samples will not affect acceptance of the lot, but may indicate future mandatory action.

6.2.3 Data requirement.- The following data are required by this specification to be furnished to the Government and should be entered on DD Form 1423 if made a part of the contract:

First article inspection
Quality conformance inspection results
Contractor furnished inspection equipment designs.

6.2.4 Level of packaging.- Packaging (level A or B) shall be a contract provision.

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